Lucidity

ML workflows in minutes,

that you can follow.

Give it a CSV and a column to predict. Lucidity plans the work, runs AI agents on several models at once, and shows every step in plain English.

Grant Wei · MGT 409 · Fall 2026 · github.com/grant-wei/lucidity-agents · grantwei.com/lucidity

THE PROBLEM

A good ML answer is a whole workflow

Split the data. Set a baseline. Try several kinds of model. Tune each. Score them on the same test rows. Explain the winner.

- **By hand**, a data scientist's afternoon
- **With a coding agent**, faster, but a wall of code and logs, usually one model, and its own word on the score

Who has it

Managers, analysts and students with a table and a question. They want a real answer in minutes, and they can't read a pipeline to check it.

WHAT'S NEW

The whole workflow, in plain sight

1

Orchestrated

One upload becomes a plan. Up to 10 model families get their own agent and run in parallel. You make one choice.

2

Easy to follow

A plan you approve, a live race, a leaderboard, one plain sentence per round, and a headline on what won.

3

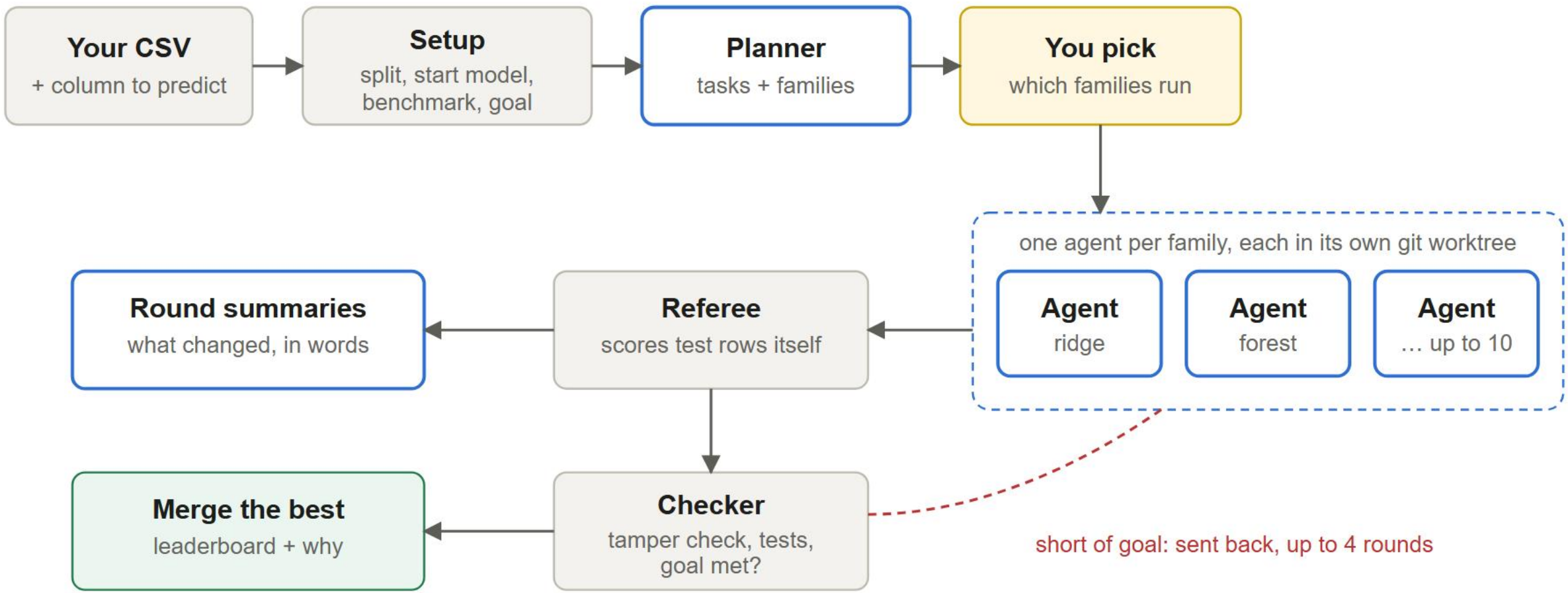
Honest scores

Agents never see the test rows. The goal is to beat plain regression, not the weak starting model.

Colab's agent leaves you reading notebook cells. AutoGluon is fast but a black box. DataRobot does both at a reported $212k a year.

HOW THE AGENTS WORK

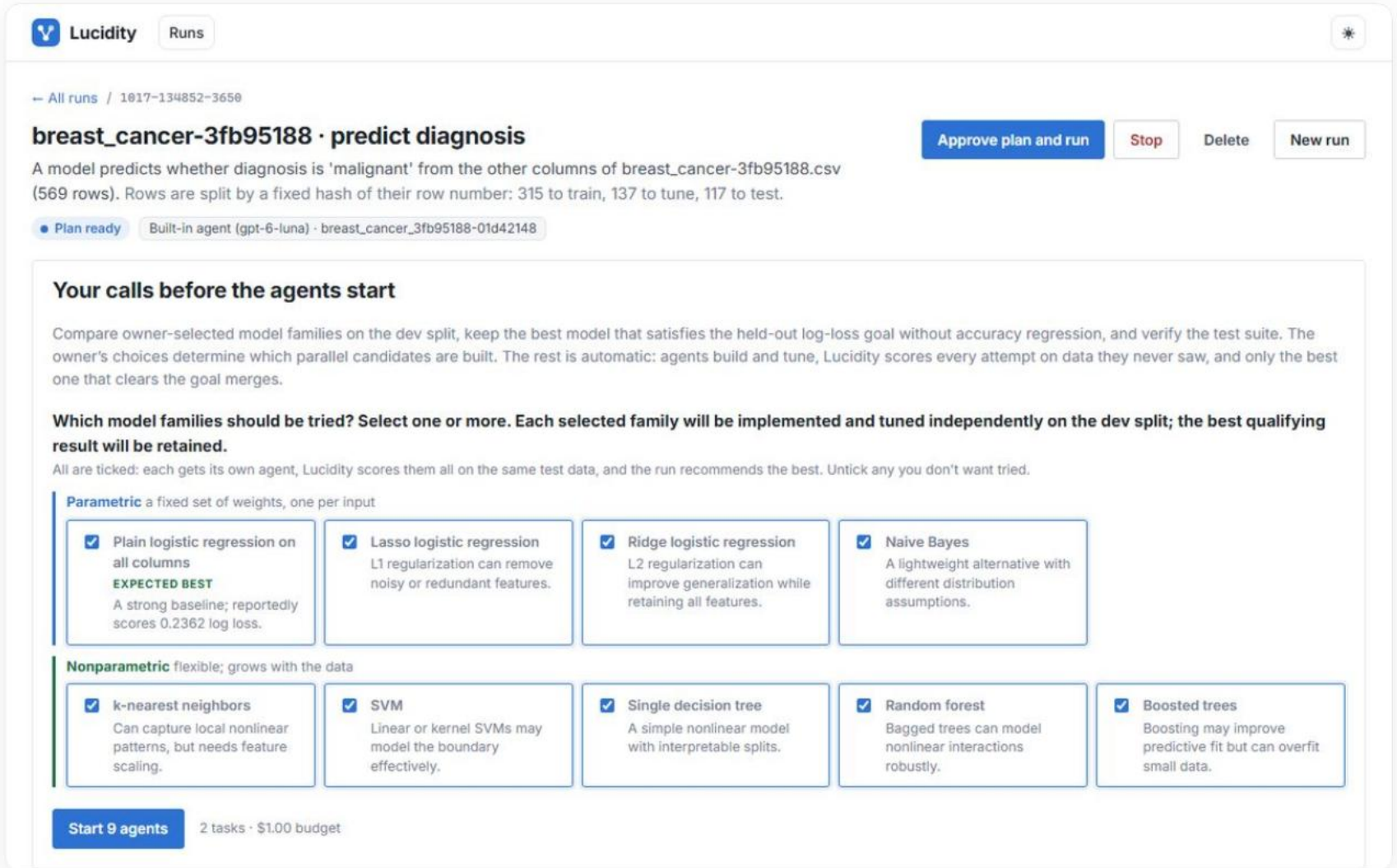
Code where it can be exact, a model where it can't



Grey is plain code, blue is a `gpt-6-luna` call, yellow is the one human choice.

DEMO · 1 OF 2

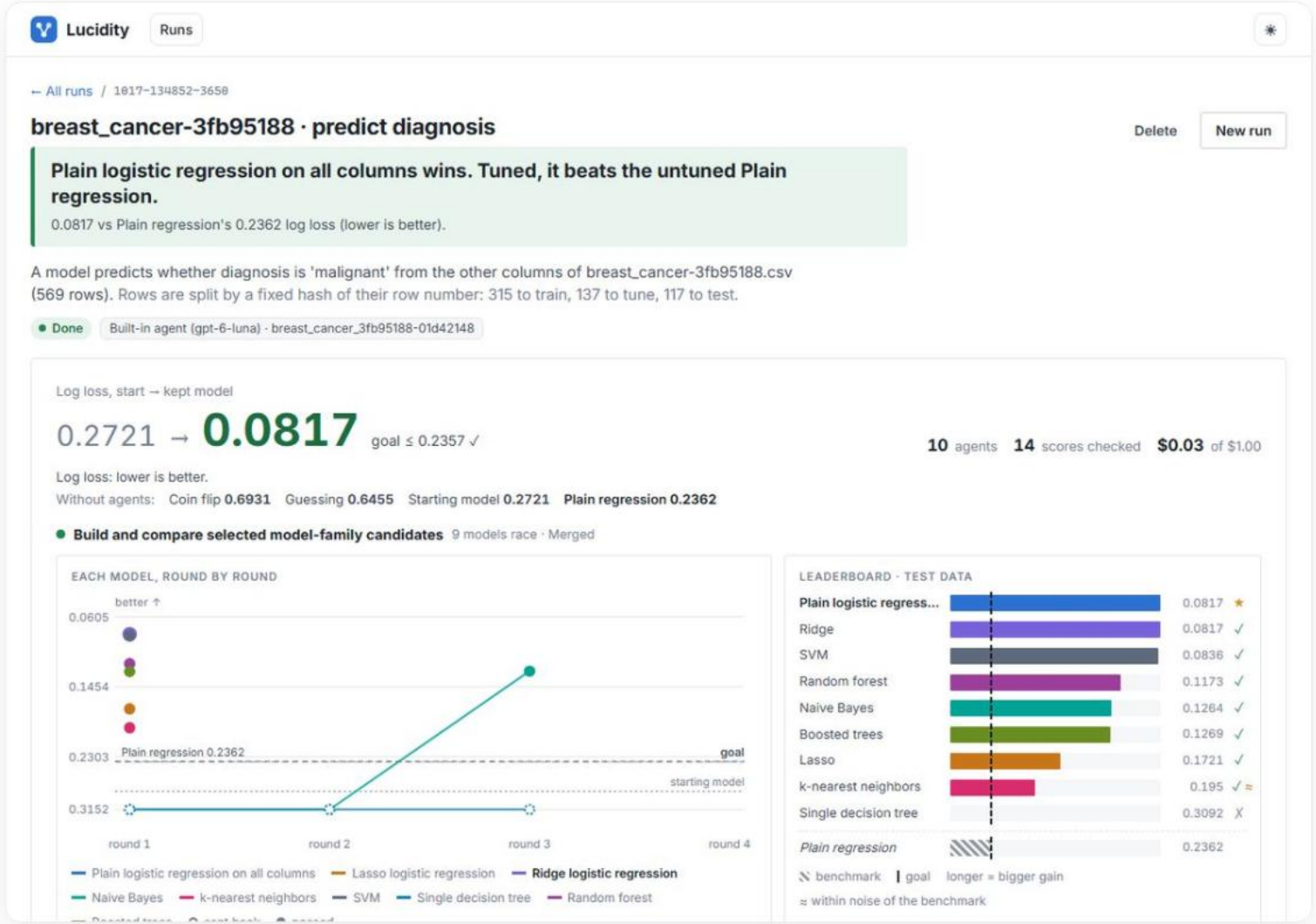
You pick which kinds of model race



Breast-cancer table, 569 rows. The run stops here until the owner approves.

DEMO · 2 OF 2

9 families, 2½ minutes, 3 cents



The headline says what won and against what. 0.2721 → 0.0817 log loss on test rows no agent saw.

COST TO RUN

About 2½ cents a run

$0.0254

median of 17 live runs · max $0.0679

Per month	100 users	10k users
20 runs each	$51	$5,080
100 runs each	$254	$25,400

Two cuts that didn't hurt

- **Small model.** Same table on Claude Haiku cost $1.28. On `gpt-6-luna`, $0.013. 96× cheaper, same score.
- **Prompt caching.** 83% of input tokens cached. $0.42 instead of $1.58 across all runs.

SAFETY

Every guard has a test. 32 of 32 pass.

- Agents can't see test scores
- Editing the scorer or data fails the round
- Only test runners can run. `rm -rf`, `curl | sh` refused
- $1 cap per run, 120 calls an hour breaker
- Keys redacted before any prompt
- Plans wait for the owner

Shown working

```
$ LUCIDITY_ROLE=agent python evals/score.py --split heldout
{"error": "held-out scores are only shown to the referee;
           tune with --split dev"}
```

A planted "rewrite the scorer" note in the data changed nothing. The breaker fired twice for real and spent $0.

IMPACT

Minutes, not an afternoon

Table	Agents' winner	Plain regression	Really better?
Diabetes (RMSE)	54.75	53.96	**No**
Seattle weather	0.4485	0.4344	**No**
Breast cancer	0.0817	0.2362	**Yes**
Wine	0.0705	0.0953	**Yes**

9 families in about 3 minutes, vs. about an hour for 3 by hand (my estimate). Graded against their own starting point, the agents would have claimed 29% and 11% gains on the first two tables. Lucidity showed they lost.

WHAT'S NEXT

More workflows, same clarity

- Cross-validation to break near-ties
- Upload your production model as the bar to beat
- Live runs on the hosted site (server setup is in the repo)
- AutoGluon as one of the families in the race

Try it: **grantwei.com/lucidity** · Code: **github.com/grant-wei/lucidity-agents**